Multi-Phase DARTS-Based NAS with NEPS Optimization

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Modality 2/3

- [x] Week 1
- [x] Week 2
- [x] Week 3
- [ ] Week 4
- [ ] Week 5
- [x] Week 6
- [ ] Week 7
- [ ] Week 8
- [ ] Week 9
- [ ] Week 10
- [ ] Bonus
- [ ] Literature

INTRODUCTION

- Neural Architecture Search (NAS) aims to automate the design of deep learning models.
- DARTS is a popular NAS method that enables efficient search through continuous relaxation.
- NEPS provides a flexible framework to run and evaluate NAS experiments seamlessly.

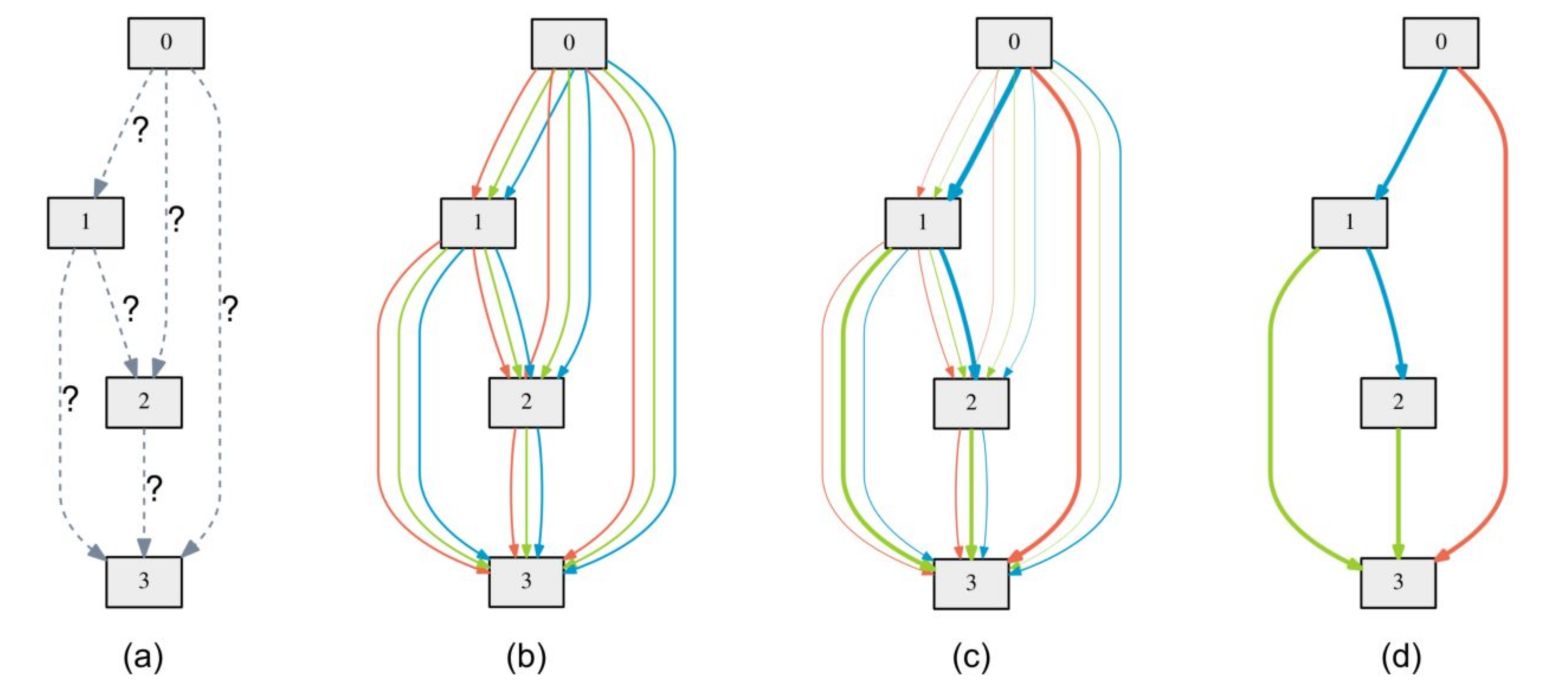


Fig 1. Darts Architecture

DARTS

- relaxes discrete architecture choices into a continuous space using softmax weights.
- uses bi-level optimization to separately update network weights and architecture parameters.
- Gradient descent is used for efficient and scalable architecture search.
- A small cell is searched and then stacked to build the final network architecture.

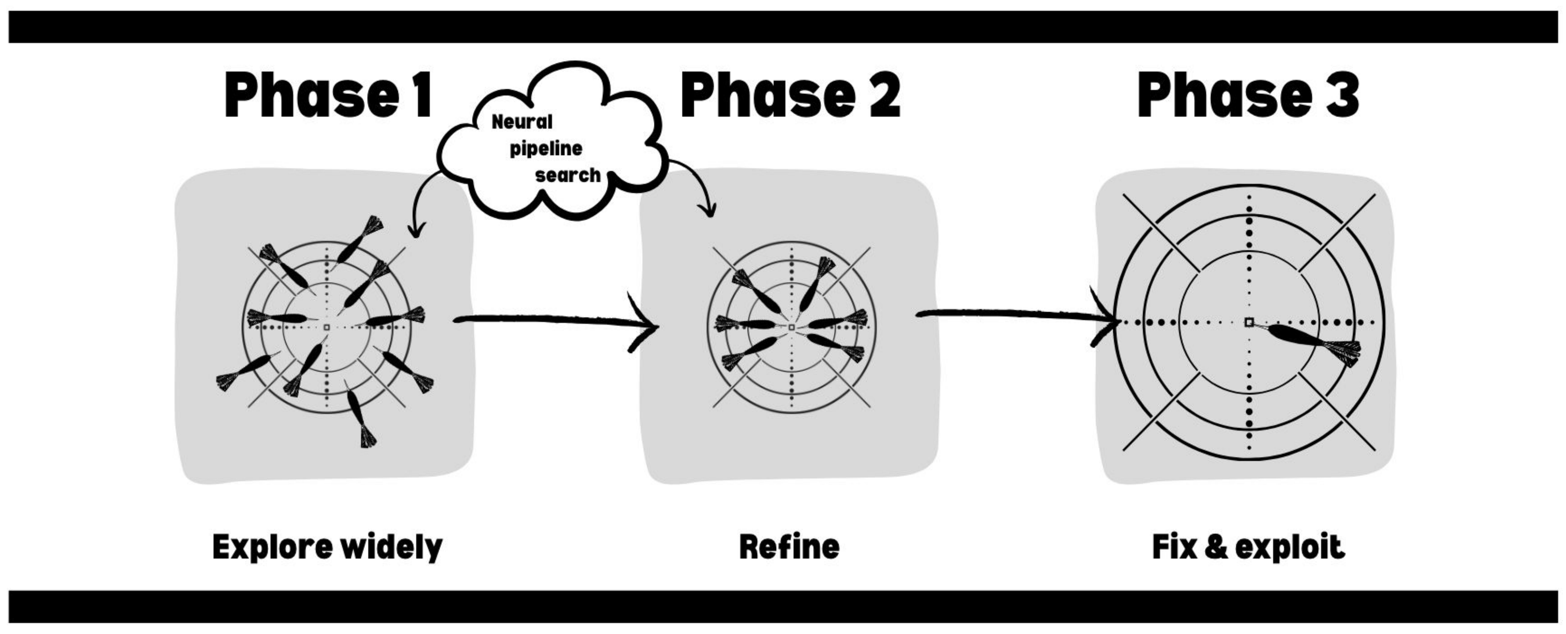


Fig 2. Interpretation of our pipeline

METHOD

- **Phase 1:** Use proxy model with 3 combined training datasets; perform wide-range hyperparameter search using NEPS.
- **Phase 2:** Load best proxy model from Phase 1; perform narrow-range NEPS search on final dataset to refine hyperparameters.
- **Phase 3**: Extract best genotype from Phase 2; fix architecture, scale it, and train fully.

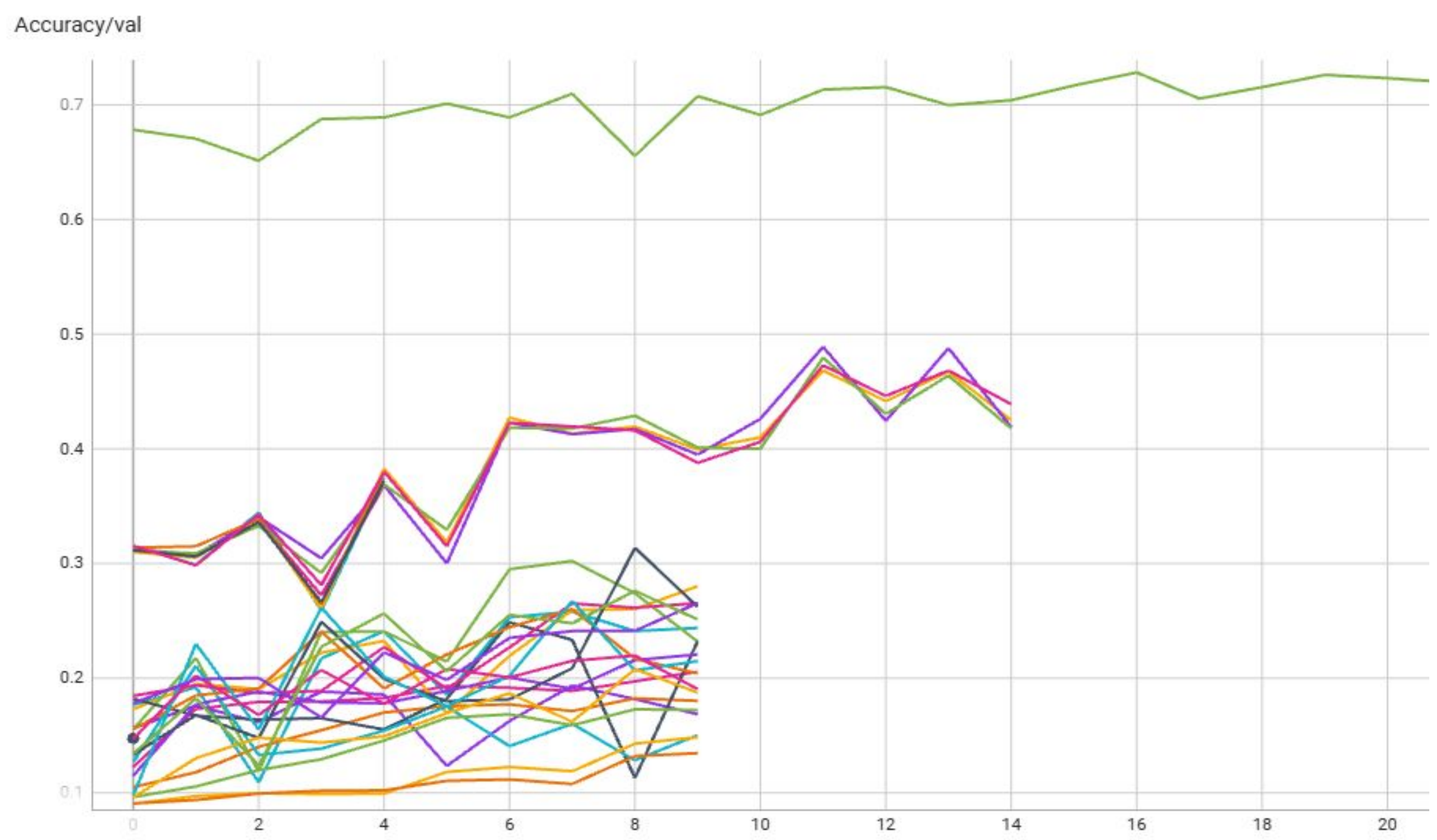


Fig 2. Interpretation of our pipeline

Resources Used

For development:
- 1 RTX 3050 6GB GPU
- 1 GTX 1650 4GB GPU
- 12 - 15 hr approx.

For AutoML:
- 1 RTX 3050 6GB GPU
- 9 hr

Workforce:
- 1 full week on average

Experimental Setup

Proxy Model:
- Width: 12
- Layers: 4
- Max batches per epoch: 100

Final Model:
- Width: 32
- Layers: 15

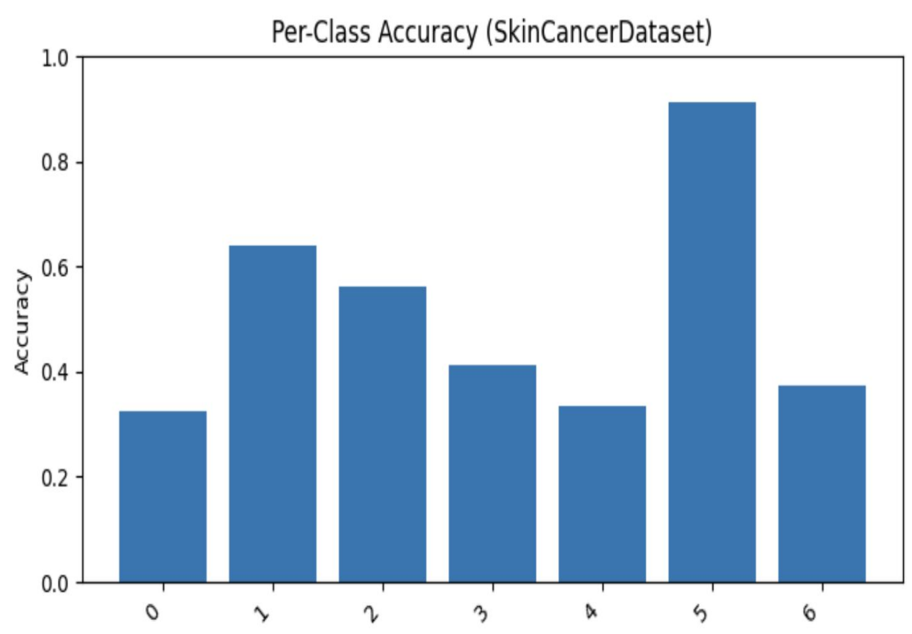
Training:
- Max input dimension: 96
- Batch size: 18
- Optimizer: Adam
- Loss: Cross-Entropy

NePs:
- Learning rate: Alpha, weights
- Gradient clip
- Weight decay
- Optimizer: Hyperband

Result & Conclusion

- The full pipeline (Phases 1–3) yields strong validation and test accuracy on all datasets (see Table).
- Skipping Phase 1 leads to slower convergence and lower accuracy, even with the same training time.
- Phase 1 helps identify strong hyperparameters early, accelerating downstream training and improving final results.
- Phase 2 remains important for fine-tuning on target datasets, especially those that are dissimilar to the Phase 1 training set.

Datasets	Validation Accuracy	Test Accuracy
Fashions	0.92	0.90
Emotions	0.63	0.61
Flowers	0.63	0.46
Skin cancer	**0.76**	**0.75**



Number of queries for test score generation: **2**

References

[1] Liu, H., Simonyan, K., & Yang, Y. (2019). DARTS: Differentiable architecture search. In International Conference on Learning Representations (ICLR).
[2] Stoll, D., Mallik, N., Schrodi, S., Janowski, M., Garibov, S., Abou Chakra, T., Rogalla, D., Bergman, E., Hvarfner, C., Binxin, R., Kober, N., Vallaeys, T., & Hutter, F. (2023). Neural Pipeline Search (NePS) (v0.11.0) [Computer software]. GitHub. https://github.com/automl/neps